ONLINE EVENT TECHNICAL CHALLENGE

Bimanual VLA Manipulation with Multi-Modal Reasoning

Challenge Option: Setting Up a Dinner Table

DIFFICULTY **Intermediate to Advanced**	ROBOT TARGET **Simulated Dual SO-101 Arms**	PRIMARY SIMULATOR **MuJoCo**
DEPLOYMENT TARGET **Intel Core Ultra Series 2/3**	INFERENCE RUNTIME **OpenVINO**	EVENT FORMAT **Online / Simulation-first**

Challenge Overview

Build an end-to-end Physical AI solution for bimanual robotic manipulation in simulation. Participants will use two simulated SO-101 arms in MuJoCo to interpret natural-language instructions, reason over camera observations, coordinate both manipulators, and complete a multi-step table-setting task. The solution should combine Vision-Language-Action (VLA) or related policy models with Intel software for model optimization and edge deployment.

Core Goal

Demonstrate a robust perception-to-action pipeline in which a multi-modal policy understands a task instruction, reasons over the simulated scene, coordinates two robot arms, and completes the requested manipulation sequence on an Intel Core Ultra Series 2/3 system.

Target Scenario: Setting Up a Dinner Table

- **Multi-task manipulation:** Open a drawer, retrieve spoons and forks, pick up a plate and cup, and organize the items on the table.
- **Dual-arm coordination:** Use both arms for hand-offs, coordinated reaching, and complementary actions such as one arm holding a mug while the other pours from a bottle.
- **Natural-language execution:** Execute commands such as: “Open the top drawer, pick up the plate with arm A, place it on the table, pick up the mug with arm B, pour water into the mug with arm A.”

TECHNICAL SCOPE

Technical Objectives

1. Bimanual Manipulation

Develop coordinated control for two SO-101 arms, including object hand-off, collision-aware sequencing, shared workspace reasoning, and multi-step task execution.

2. Multi-Modal Reasoning

Process natural-language task instructions together with raw camera observations to identify objects, infer task state, choose the next action, and maintain context across a multi-step manipulation sequence.

3. Robustness Under Perturbation

Evaluate the policy under changes in object weight, friction, shape, lighting, background, and initial object placement. The solution should maintain task success without depending on one fixed scene configuration.

4. Simulation Training and Policy Distillation

Train or fine-tune a robotics policy in MuJoCo using Hugging Face LeRobot or compatible tooling. Candidate policies may include SmolVLA, Pi0.5, ACT, or another appropriate VLA / imitation-learning policy.

5. Intel Edge Optimization

Optimize the inference pipeline for Intel Core Ultra Series 2/3. Where applicable, quantize and compile supported model components to OpenVINO IR and target Intel CPU, iGPU, and/or NPU acceleration.

Expected End-to-End Workflow

01	02	03	04	05
Observe	**Understand**	**Plan**	**Act**	**Optimize**
Camera feeds + simulator state	Language + scene reasoning	Bimanual action sequence	Dual SO-101 manipulation	OpenVINO on Intel Core Ultra

INTEL PHYSICAL AI | ONLINE CHALLENGE

intel

PLATFORM, SOFTWARE, AND DEPLOYMENT

Platform Requirements

- **Target system:** Intel Core Ultra AI PC / NUC, or a local execution environment with Intel CPU and iGPU acceleration.
- **Preferred deployment hardware:** Intel Core Ultra Series 2 or Series 3 system for final inference and demonstration.
- **Simulation:** MuJoCo, or compatible LeRobot Gym environments when appropriate for development and evaluation.
- **Training hardware:** Developers may train on any local hardware or cloud instance. Training infrastructure is not provided by Intel.

Software Resources

Intel OpenVINO Toolkit - Model conversion, optimization, quantization, and inference on Intel hardware.

OpenVINO Physical AI - Physical AI-oriented model deployment and robotics inference workflows.

Intel Physical AI Studio - Development environment for building and integrating Physical AI / VLA robotics workflows.

Intel Open Edge Platform - Edge software foundation for deploying and managing AI-enabled workloads.

Intel Edge AI Suites - Robotics - Reusable edge AI components and robotics-oriented capabilities.

Intel Geti - Computer-vision development tooling that may be used where relevant to perception workflows.

Deployment and Demonstration Requirements

- **Run the final simulation on Intel hardware:** MuJoCo and the AI/VLA/VLM inference pipeline must execute on an Intel Core Ultra Series 2/3 system for the final demonstration.
- **Optimize with OpenVINO:** Use OpenVINO model optimization where supported, with emphasis on reducing latency and improving efficient utilization of Intel CPU, iGPU, and NPU resources.
- **Preserve system behavior:** Optimization should not materially degrade the task-success rate or the quality of multi-step reasoning and manipulation.

Important

The challenge is simulation-first. Physical SO-101 hardware is not required for the online event. The target robot is a simulated dual SO-101 configuration in MuJoCo, while the AI inference and simulation workload are demonstrated on Intel client hardware.

INTEL PHYSICAL AI | ONLINE CHALLENGE

intel

SUBMISSION PACKAGE

Required Deliverables

1. Reproducible GitHub Repository

Repository containing setup instructions, environment dependencies, MuJoCo scene/assets, policy training or fine-tuning code, evaluation code, inference code, and clear commands to reproduce the demonstration.

2. Reproducible MuJoCo Simulation

A simulation package that recreates the dual-arm dinner-table scenario, including environment randomization and evaluation configuration.

3. Intel Inference Benchmark Script

A bench-test script that runs on Intel Core Ultra Series 2/3 and reports relevant model performance such as latency, throughput, device selection, and model precision.

4. Demonstration Video

A video demonstrating successful task execution across 10 randomized environment seeds. The video should make the command, scene variation, and robot outcome easy to verify.

5. Technical Readme / Architecture Summary

A concise description of the solution architecture, VLA/VLM model choice, bimanual coordination strategy, training approach, robustness methods, OpenVINO optimization, and Intel hardware mapping.

Recommended Demonstration Sequence

- Present the natural-language command and the randomized initial scene.
- Show perception / policy inference operating from simulated camera observations.
- Demonstrate coordinated actions across both SO-101 arms, including at least one hand-off or complementary dual-arm action.
- Complete the table-setting task and show final task state.
- Repeat evaluation across 10 randomized seeds and summarize the success rate.
- Run the benchmark on Intel Core Ultra Series 2/3 and report OpenVINO optimization results.

INTEL PHYSICAL AI | ONLINE CHALLENGE

intel

EVALUATION

Judging Criteria

Submissions are evaluated on a 100-point scale. The rubric emphasizes end-to-end task completion, multi-modal reasoning, bimanual coordination, robustness, reproducibility, and efficient Intel deployment.

Criteria	Description	Points (100)
End-to-End Task Completion & Bimanual Manipulation	Completes the requested dinner-table workflow in MuJoCo using two SO-101 arms. Evaluation includes sequencing, object hand-off, coordination, manipulation accuracy, and overall task success.	**30**
VLA / Multi-Modal Reasoning	Correctly interprets natural-language instructions and visual observations, maintains multi-step task context, selects appropriate actions, and adapts the plan as scene state changes.	**20**
Robustness & Generalization	Maintains performance under randomized object placement, weights, friction, shapes, lighting, and background conditions. Demonstration should include results across 10 randomized seeds.	**15**
OpenVINO & Intel Core Ultra Optimization	Demonstrates optimized inference on Intel Core Ultra Series 2/3 using OpenVINO where supported. Evaluation considers latency, throughput, precision/quantization choices, device utilization, and preservation of task quality.	**20**
Technical Quality & Reproducibility	Provides a clear, reproducible repository, deterministic setup instructions, evaluation tooling, benchmark scripts, and a well-structured MuJoCo environment.	**10**
Innovation & Technical Demonstration	Shows a thoughtful or novel approach to bimanual coordination, policy design, robustness, optimization, or system integration, and communicates the solution clearly in the final demonstration.	**5**
TOTAL		**100**